

WAYNE KIRK SCHMIDT

PROFILE

Principal Technologist, security platform builder, and engineering leader with 30+ years designing, building, and operating mission-critical systems at institutional scale, built with Security, DevOps and ML/AI as the core and foundation.

Built Production Perimeter, a patented ML-driven adaptive security enforcement platform acquired by Illumio. Built BEACON, an ML-driven compromise detection system reducing identification time by 97% across large fleets. Built ARK, an ML-driven dependency management platform across 60,000+ hosts.

Led engineering and market data teams of up to 30 across Morgan Stanley and Nomura, including a \$107M portfolio spanning 7 APAC markets and perimeter engineering across 80+ institutional trading locations.

Hands-on background in zero-trust architecture, threat detection, distributed systems, and AI-driven security automation, combined with executive ownership of Tier-0 systems under adversarial and regulatory pressure.

Security Engineering · Zero-Trust Architecture · ML-Driven Threat Detection · Distributed Systems · Multi-Cloud Infrastructure · Auth & Identity · Platform Engineering · AI/ML in Production

EXECUTION	LEADERSHIP	ADVISORY
Distributed, low-latency, AI-enabled systems at scale. Hands-on depth across DevOps, ML/AI, and Security from 60K+ host fleets to ULL data platforms.	Engineering and operations ownership across Tier-0 systems and \$107M+ portfolios. Teams up to 30 engineers across global and regional operations.	Supported 60+ clients across 6 industries and 4 regions, allowing SME, MSSP, and Enterprise to navigate platform risk, security, operations, and transformation.

SIGNATURE OUTCOMES

REALIZED	GUIDED	DELIVERED
\$300M+ measured impact across products, services, and portfolios in AMER and APAC.	5 zero-to-one production-grade platforms across security, finance, and operations.	90+ initiatives in system stabilization, digital transformation, and operational excellence.

- Invented **Production Perimeter**, an ML-driven security platform deployed globally: patented IP later acquired by Illumio.
- Delivered **NXT RAW**, a ¼µs ultra-low latency market data platform generating **\$14M+** in revenue supporting multiple APAC markets.
- Led a **\$107M** market data portfolio over **7 APAC** markets, **30-person** team, **15 vendors**, consistent sub-millisecond performance.
- Built **ARK**, ML-driven dependency platform for **60,000+** hosts, reducing deployment risk by **97%**, speeding releases **17×**.
- Redesigned event management for US MSSP, reducing response from **20 minutes** to **10 seconds** at **1.5 million events** per minute.
- Delivered Jira/Power Automate AI workflows for a LATAM bank, cutting incident management from **2 person-weeks** to **15 minutes**.

Detailed Delivery Case Studies: casestudies.changeis.io

PROFESSIONAL EXPERIENCE

ChangeIS	Founder & Principal Technologist	Tokyo, Japan	2013 – Present
Independent advisory and engineering practice in ML/AI, DevOps, and security orchestration across finance, insurance, and technology globally.			
- Built a Kafka-Logstash-Grafana observability platform for a US MSSP, reducing incident response time by 99.16%. - Deployed ML-driven log analysis platform reducing issue triage effort by 85% and reducing MTTR by 3×. - Created ML-driven asset platform for global insurance firm, raising accuracy to 99.9%, reducing reporting lag by 99.93%. - Administered and tuned PostgreSQL database infrastructure for a high-touch equity trading client, managing replication and failover to sustain low-latency, high-availability performance under production trading load. - Built Snowflake-based data warehousing and Spark-driven data pipelines supporting analytics, reporting, and downstream ML workloads across client engagements.			
Insight Innovation Fund	Head of Product (Fractional / NED)	Tokyo, Japan	2024 – Present
- Patented MORPHOS/9STACK®, a distributed reasoning-based AI orchestration framework for private, adaptive, edge deployment. - Directed product architecture and enterprise AI solutions, advising on go-to-market, partnerships, and M&A strategy.			
Sumo Logic Japan, K.K.	Senior Solution Engineer	Tokyo, Japan	2018 – 2022
- Built 10+ vendor integrations across security and DevOps, expanding platform addressability for enterprise clients. - Developed automated monitoring tooling reducing deployment time by 82%, adopted as the APAC reference tooling. - Authored 40+ client solutions across DevOps and observability, supporting renewals and expansion revenue in Japan.			

Tanium G.K. | Executive Director – Technical Account Management | Tokyo, Japan | 2015 – 2016

- Created **BEACON**®, an ML-driven compromise threat detection system, reducing compromise identification time by **97%**.
- Built **BRAVO**®, an ML-driven testing framework saving an estimated **\$1.8M** in engineering effort.
- Architected compliance reporting for **1,000+** Fuji Xerox printers via Tanium, shortening audit/reporting cycles from weeks to days.

Smart Technology Partners | Chief Technology Officer (Fractional) | Japan | 2013 – 2016

- Designed a green energy cloud architecture using refurbished hardware, delivering sustainable infrastructure for 10+ clients.
- Delivered Digital Transformation proposals for regional government clients across Fukushima and Kumamoto prefectures.
- Developed cloud solutions across logistics, genetic data processing, and data center security.

Nomura Securities | Executive Director – Asia Head of Market Data / Global Head of Market Data Engineering | Tokyo, Japan | 2009 – 2013

- Managed a **\$107M** portfolio over **7 APAC** markets, **30-person** team, and **15 global vendors** supporting trading and retail functions.
- Delivered **NXT RAW**, a $\frac{1}{4}\mu\text{s}$ ML-driven TSE data platform generating **\$14M+** in revenue.
- Built a DevOps automation framework reducing engineering effort by **70%** across a **\$25M** project portfolio.

Morgan Stanley Securities Japan | Vice President – Infrastructure Architect, Asia-Pacific | Tokyo, Japan | 2003 – 2009

- Designed **ARK**, ML-driven dependency management supporting hosts, reducing deployment risk by **97%** and releases by **17×**.
- Delivered **Production Perimeter**, a patented self-learning ML firewall preventing hundreds of millions USD in potential trading losses.
- Rebuilt Asia Equities QA and Integration Testing Environment, resolving systemic regional stability issues, increasing reliability to **99.93%**.

Morgan Stanley | Vice President – Perimeter Security Engineering | New York, New York | 1998 – 2003

- Built **AEGIS**, cross-platform security and configuration management across 80+ institutional trading locations.
- Built **ARGUS**, a firewall lifecycle automation suite coordinating global plant builds and reducing misconfiguration risk.
- Led the firm's Y2K global security infrastructure refresh, replacing at-risk legacy systems across global operations.

EARLY CAREER: BOSTON & NEW YORK

Systems and infrastructure roles establishing technical foundations in networking, distributed systems, and enterprise platforms:

- GTE Internetworking Systems: automated provisioning across **500+** servers, improving restore performance by **20×**.
- Wildfire Communications: led 4-person sysadmin team, automating Windows NT provisioning from **7 hours** to **20 minutes**.
- Pencom / Collective Technologies: architected Tivoli-based software deployment platform for a **5,000**-laptop fleet.
- Salomon Brothers: fixed income trading support, delivered **15×** reduction in risk calculation time, **\$1.3M** procurement savings.

PRODUCTS & PATENTS

MORPHOS/9STACK®	Reasoning-based AI orchestration; private, adaptive, edge-ready, with memory, context, and bounded ethics.
Production Perimeter®	ML-driven adaptive firewall; self-learning security enforcement across LAN, MAN, WAN. Acquired by Illumio.
NXT RAW® / NXT	Ultra-low latency market data delivery; sub- $\frac{1}{4}$ microsecond signal delivery at institutional scale.
BEACON®	ML-driven threat detection system; 97% reduction in compromise identification time across endpoint fleets.

Additional patents and products in event management, operational orchestration, security automation, and distributed systems architecture.

SELECTED TECHNICAL SKILLS

DevOps / Automation	Python, Bash, PowerShell, Ansible, Terraform, Puppet, Chef, Kubernetes, Docker, VMware, Jenkins, GitHub Actions, GitLab CI, AWS, GCP, Azure, OpenStack, Kafka, RabbitMQ, Tibco, PostgreSQL, Snowflake, Apache Spark (Databricks-compatible), Splunk, Sumo Logic, Elastic Stack, Grafana, Prometheus, Nagios
ML / AI Systems	scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, LangChain, spacy, NumPy, Pandas, Reasoning architecture design, Adaptive workflow systems, Low-latency ML inference, Edge & federated deployment, MORPHOS/9STACK® (patented), Matplotlib, Plotly, Dash
Security / Resilience	Tanium, CrowdStrike, Auth0, Recorded Future, Zeek, eBPF, Zero-trust architecture, NIST 800, ISO 27000, COBIT, SOX, ISO 31000, Checkpoint, Cisco (IGP/EGP routing, NGFW deployment & management), Arista, Palo Alto, RDMA, ML-driven threat detection, Compliance & audit automation, Wireshark, Rkhunter

Additional Capabilities: waynekirkschmidt.me/technical

EDUCATION & LANGUAGES

Ohio State University — B.S. Physics (dual study: Physics & Fine Arts)

Adjunct Professor, Continuing Education — New York University · Programming, Networking, and Security

English: Native Japanese: Professional working proficiency · JLPT N3 · N2 in progress